

Agentic AI and Managers' Analytics Capabilities: An Exploration

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Abstract

As firms across industries adopt data-driven decision-making practices, the need for managers fluent in advanced analytics has grown increasingly urgent. Business schools have responded by expanding advanced analytics offerings in MBA programs, but the wide range of student backgrounds—spanning “poets and quants”—poses challenges for uniformly developing these capabilities. Can agentic-AI tools help bridge this gap? We investigate this question through a field deployment of a no-code, agentic-AI tool—functioning as a virtual data scientist—in a core analytics course at a top-30 global MBA program. This tool allows students to solve machine learning problems using natural language across a range of tasks, including segmentation, classification, prediction, and text analytics. Using detailed logs of student-agent interactions and closed-book final exam scores, we examine (1) whether use of the agentic-AI agent improves learning outcomes, (2) whether usage patterns differ across students, and whether these differences affect learning, and (3) what mechanisms link agentic-AI usage with learning outcomes. We find that doubling the intensity of agentic-AI usage increases a student’s final exam score by an average of 0.624 points (or about 1.56 percentage points)—exceeding the typical gap between an ‘A’ and an ‘A-.’ Lower-performing students are less likely to use the agent, but when they do engage, they experience greater marginal returns. These performance gains are concentrated in the applied business case portion of the exam, rather than the analytics concepts section. Notably, students benefit most when they adopt a “going-to-the-gym” usage mode—actively working through MBA-level case exercises with the tool—rather than relying on it for exam preparation or passively observing teammates.

Keywords: Agentic AI, Generative AI (GenAI) in Education, Analytics Skill Development, Data-Driven Decision-Making (DDD), AI-Augmented Learning, MBA Education, Educational Equity, Digital Learning Behavior

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1. Introduction

Nearly three decades of investments in enterprise systems like ERP and CRM, followed by the dotcom boom and the smartphone revolution—which captures atomic-level data on everything from dating and gaming to everyday social interactions—have together created an ever-expanding ocean of data. It is widely acknowledged that harnessing this data through the full spectrum of advanced analytics—pursuing data-driven, evidence-based scientific decision making—can significantly enhance firm performance. In fact, companies that effectively leverage multiple components of what Bapna and Ghose (2024) term the “House of AI”—which includes descriptive, predictive, causal, and prescriptive analytics—embody what McAfee (2023) calls “The Geek Way.” These firms consistently outperform their industry peers. Brynjolfsson et al. (2011), for example, find that firms adopting data-driven decision-making (DDD) exhibit output and productivity levels that are 5-6% higher than expected. Yet, despite their outsized visibility in the media, such ‘Geek Way’ firms are rare, and the adoption of DDD is uneven (Brynjolfsson and McElheran 2016, McElheran et al. 2024). Indeed, industry reports¹ suggest that companies currently utilize only a single-digit percentage of the data they generate for insight generation.

Bapna and Ghose (2024) emphasize that the impetus to harness the full scope of the ‘House of AI’ must come from leadership and management. Success in DDD, they argue, depends on close synergistic collaboration between managers and data scientists. Yet as Malone (2020) observes, too many managers remain at arm’s length from data scientists, failing to engage in the hard work of defining core problems at hand and clearly articulating the measurable outcomes they hope to achieve. Addressing this gap requires more than organizational alignment; it calls for managerial fluency in the AI and analytics landscape. To identify and shape high-value projects that deliver tangible business results, managers must be familiar with both the foundational pillars of analytics (descriptive, predictive, causal, and prescriptive) as well as the key advanced layers (reinforcement learning, deep learning, generative-AI (Gen-AI) and explainable and fair use of AI).

¹ <https://aws.amazon.com/executive-insights/content/the-power-of-the-data-driven-enterprise/>

Not surprisingly, business schools have taken seriously the need to equip MBA students—managers in training—with analytics skills, expanding course offerings across their programs. While analytics electives have existed in some curricula for decades, there is a nascent—and to our knowledge still unquantified—movement towards making analytics a required core course.² This shift, however, poses challenges: many MBA students often self-identify as “poets” rather than “quants” and tend to shy away from machine learning and data science-oriented courses that are associated with coding and perceived as being math-intensive.

As business schools navigate the challenge of broadening analytics education, can Gen-AI come to the rescue?

One of Gen-AI’s superpowers is its ability to write code and perform advanced analytics (Bapna and Ghose 2024). This raises an intriguing possibility: could a thoughtful deployment of Gen-AI help lower technical barriers and make analytics more accessible to a broader segment of MBA students? More specifically, we ask whether modern advances in agentic-AI can help address this challenge. We define agentic-AI as a Gen-AI actor that has agency to make decisions—in our context, this refers to decisions regarding data-science machine learning tasks that would otherwise be made by human data scientists.

To be precise, we design and deploy—at scale (425 students)—a bespoke no-code agentic-AI solution in the form of a virtual data scientist, embedded within a high-stakes core analytics course at a top-30 global MBA program. The course is considered high stakes because students’ core CGPA serves as a screening filter for first-round interview consideration by top consulting firms recruiting at the campus. This virtual data scientist allows MBA students to use natural language—English rather than R or Python—to tackle machine learning cases that span a range of techniques: from unsupervised learning (e.g., segmenting customers at a bank) to binary classification (e.g., detecting cancer) to numeric prediction (how much will a shopper spend) and natural language processing (e.g., detecting fake reviews). This context—

² Getting a new course into the MBA core is tall undertaking that is rife with turf wars between departments. The trend is to shrink the core curriculum and expand the choice of electives. For a new course to become core, some other course has to not be core.

where MBA students interact with an agentic-AI tool in the form of a virtual data scientist—allows us to examine the following high-level research questions:

1. Can an agentic-AI-based virtual data scientist help MBAs develop stronger analytics capabilities?
2. Is there heterogeneity in the usage of the agentic-AI across key student characteristics—for example, between lower- and higher-performing students—and does this heterogeneity differentially impact learning?
3. What are the specific mechanisms that link agentic-AI usage with learning outcomes?

While our first research question follows from the motivation to broaden analytics education among MBA students, the second and third questions speak to behavioral dynamics underlying agentic-AI use. Specifically understanding how usage varies across students—and whether these variations translate into different learning gains—is particularly important in graduate-level business education, where learners vary in their ability and comfort with digital technologies. Moreover, knowing *whether* agentic-AI helps is just part of the picture; understanding *for whom, under what conditions, and how* it helps is essential for designing effective educational interventions. These factors round out the high-level motivations behind our three research questions.

To answer them, we analyze prompt-level data and exam performance metrics collected from the aforementioned core MBA analytics course. To address our first research question, we model the relationship between usage intensity and learning, as captured by performance on a proctored, in-person, closed-book, pen-and-paper final exam that assesses both objective knowledge (analytics concepts) and the ability to apply data analytics techniques to solve business problems (application of concepts).³ We identify our effect based on falsification tests and other key controls for potential confounds. We find that doubling the intensity of agentic-AI usage increases the average student’s final exam score by 1.5% points—a meaningful gain, exceeding the typical difference between an ‘A’ and a ‘A-’ letter grade. For

³ Closed book exam in our contest refers to an exam taken without the agentic-AI agent or any other form of assistance.

our second research question, we uncover a compelling paradox: lower-performing students, despite engaging with agentic-AI less frequently, gain more when they do use it. This suggests a compensatory effect, where agentic-AI use can help narrow performance gaps. Ironically, the very students who stand to benefit the most appear the least likely to do so, pointing to an emerging equity challenge in AI-enhanced education. Finally, with respect to mechanisms, we find that the gains in performance are concentrated in the section of the exam that evaluates application of concepts to solve business problems, rather than in questions assessing analytics concepts. In additional analyses, we show that students benefit most when they use agentic-AI in a ‘going-to-the-gym’ mode—that is, by actively doing the MBA-level case work rather than using the tool for exam preparation or passively observing team members.

Our study makes several contributions. First, we move beyond binary “access vs. no access” designs by modeling agentic-AI engagement as a continuous variable based on prompt-level usage. Second, we reconcile conflicting findings on agentic-AI’s equity effects by testing both whether lower-performing students use agentic-AI less often (usage differences) and whether they benefit more or less from it when they do use it (performance-contingent returns). Third, we study agentic-AI in a high-stakes, professional-skills context—a required MBA analytics course tied to job placement outcomes. Fourth, we distinguish between types of agentic-AI use and show that active case-solving yields the greatest learning gains. Fifth, we assess durable knowledge by measuring performance on a closed-book, proctored exam without GenAI access. We elaborate on these contributions in the “Discussion” section of the manuscript.

In the next section, we develop our theory-based hypotheses that guide our subsequent empirical analyses.

2. Theory and Hypotheses

Building on the motivation and background, we now articulate the mechanisms that link Gen-AI engagement to learning outcomes and derive testable hypotheses.

2.1. Intensity of Gen-AI Use and Learning Outcomes

Graduate business education in DDD prepares future managers to leverage data when considering organizational choices. DDD coursework demands that students dissect messy datasets, interpret statistical evidence, and articulate defensible choices—tasks that reward both analytic rigor and conceptual precision. In this high-stakes setting, Gen-AI agents—chat interfaces powered by large language models—have emerged as informal tutors because they can clarify technical concepts, critique code, and illustrate decision scenarios the moment confusion arises. Although these tools are generally viewed as productivity aids (Brynjolfsson et al. 2025, Dell’Acqua et al. 2023, Noy and Zhang 2023), we contend that the frequency with which students turn to them—i.e., the raw count of prompts submitted across a semester serves as a behavioral trace of sustained cognitive investment rather than idle curiosity. Educational-psychology research has long shown that learning is deepest when students engage in self-regulated learning loops—cycles of question generation, hypothesis testing, and feedback integration—that sharpen comprehension, strengthen memory, and cultivate intrinsic motivation (Chi and Wylie 2014, Deci and Ryan 2000). Prompting an AI agent initiates exactly this cycle. This process begins when students recognize a gap in their understanding and formulate a specific question. In interpreting the AI’s response, they evaluate whether the explanation aligns with or challenges their prior knowledge. Based on this comparison, they decide whether confusion has been resolved or whether follow-up clarification is needed, refining their question or adjusting their strategy.

These actions engage metacognitive reflections—monitoring, error diagnosis, and learning strategy revision—that decades of research link to superior academic outcomes (Chi et al. 1989, Chi et al. 1994). In effect, prompting Gen-AI agents enables compact cycles of deliberate practice: learners pursue goals (e.g., solve a problem), receive targeted feedback, revise their approach, and repeat. Higher prompt counts,

therefore, signal more frequent rehearsal of expert-like learning behaviors (Deng et al. 2024)—specifically, the planning-monitoring-reflection loop characteristic of expert learners.

Beyond enabling metacognition, Gen-AI tools function as adaptive tutors that personalize scaffolding in real time (Dell'Acqua et al. 2023, Noy and Zhang 2023). This approach aligns with learning analytics principles, which emphasize the importance of immediate, data-driven feedback in dynamic learning environments (Roll and Wylie 2016). Because these agents retain conversational context, they can detect lingering misconceptions and adjust the granularity of their explanations. This process effectively breaks down complex ideas into smaller, more manageable steps, reducing intrinsic cognitive load (Sweller 1988). The agent's ability to track prompt history and evolve its guidance supports both immediate problem-solving and long-term retention. Prompt-response cycles correct misconceptions early and prevent the reinforcement of incorrect understanding—a key limitation of traditional self-study. In this way, Gen-AI systems approach the efficacy of Bloom's (1984) "2 sigma" effect observed in one-on-one tutoring scenarios. A meta-analysis of fifty-one controlled studies reports large aggregate gains in learning performance when students use ChatGPT, with the strongest effects appearing in problem-based courses that require repeated, inquiry-driven prompting (Wang and Fan 2025). This finding suggests that high prompt frequency amplifies the individualized tutoring qualities of Gen-AI, translating interaction volume into measurable academic advantage.

Prompt frequency reflects more than simple usage—it captures how often a learner engages in feedback cycles that reinforce understanding, introduce new perspectives, and support elaborative thinking. Higher prompt frequency is associated with deeper conceptual understanding and greater knowledge transfer, as students engage more actively with feedback cycles and integrate insights over time. Research on AI-based writing assistants, intelligent tutoring systems, and coding copilots shows that academic outcomes improve most when students generate original inputs and iteratively refine outputs (Holstein et al. 2019, Clark 2024). Beyond cognitive scaffolding, Gen-AI-mediated dialogue may also encourage learners to triangulate explanations—by comparing AI feedback with peer or instructor input—which reinforces learning through multiple cognitive frames and deepens inquiry.

Taken together, these mechanisms—metacognitive engagement, adaptive feedback, and reinforcement that deepens conceptual understanding—provide a theoretical rationale for our hypothesis. We, therefore propose:

H1: Greater agentic-AI usage during a course is positively associated with learning (as measured by performance on the closed-book course final exam).

2.2. Performance-Based Differences in Gen-AI Engagement

Gen-AI agents supply students with anonymous, on-demand explanations, worked examples, and coding hints. Yet, “second-level” or “digital divide 2.0” studies show that equal access seldom translates into equal use; psychological confidence, cognitive skill, and behavioral inertia become decisive (Selwyn 2009, Van Dijk 2005, Warschauer 2004, Wei et al. 2011). Applying this perspective to graduate DDD courses suggests that students may differ in how intensively they query these systems (Hargittai 2010). Those who arrive with weaker academic records and hence lower analytic self-confidence may approach Gen-AI far more cautiously than their higher-performing classmates, submitting fewer prompts and forgoing iterative feedback when it could help most.

Lower-performing students frequently report low academic self-efficacy and doubts about their ability to master quantitative tasks. This lack of confidence discourages the trial-and-error exploration that productive Gen-AI use requires (Bandura 1977). The psychological barriers extend beyond individual self-doubt to include social-psychological considerations. Many learners anticipate “digital embarrassment” if their prompts might become visible to peers or instructors, further suppressing authentic help-seeking (Choi et al. 2024). Although Gen-AI agents offer judgment-free support, the internalized stigma of “needing help” can still deter usage. For many struggling students, simply admitting confusion—even to a machine—feels like a threat to their academic competence.

Even when students overcome initial hesitation, cognitive and metacognitive challenges can still limit sustained Gen-AI use. Prompting a Gen-AI agent is cognitively demanding: learners must diagnose what they do not understand, compress that gap into a clear question, parse a potentially complex reply, and decide on follow-up steps. For students still building basic schemas, this extra processing inflates intrinsic

cognitive load and can render early interactions confusing rather than clarifying (Sweller 1988). This cognitive burden is compounded by the metacognitive demands of effective Gen-AI use. Sustained engagement requires learners to monitor their comprehension, persist through difficulty, and strategically re-prompt routines that struggling students enact less reliably (Zimmerman 2002, Newman 1994). The situation is further complicated by metacognitive miscalibration. Lower-performing learners often overestimate how well they grasp course material and therefore assume they need little outside help, a pattern consistent with the Dunning-Kruger effect (Dunning 2011; Prather et al. 2024). This creates a problematic cycle: when early queries feel unrewarding due to cognitive overload, and self-assessment incorrectly signals "I've got this," rational avoidance follows—*I didn't get much out of it, and I don't really need it anyway*. Students curtail the very interactions that could correct their misunderstandings (Turel and Kalhan 2023). This avoidance pattern is reinforced by learned helplessness behaviors such as self-handicapping or premature withdrawal from challenging tasks (Seligman 1972, Karabenick and Knapp 1991). Ultimately, these cognitive and metacognitive challenges transform what should be a supportive learning tool into an additional source of stress.

Beyond these cognitive frictions, a status-quo bias nudges overstretched students toward familiar, low-effort study routines, making tools like Gen-AI seem not worth the switch (Samuelson and Zeckhauser 1988). Higher-performing students—with greater digital agency—by contrast, recognize knowledge gaps more accurately and treat Gen-AI as a productivity enhancer and learning accelerator. Early evidence accords with this logic: students with higher prior achievement are more likely to experiment with Gen-AI tools and iterate on feedback, while their lower-performing peers remain under-engaged despite equal access (Bastani et al. 2024, Nie et al. 2024).

Taken together, low self-efficacy, heavy cognitive and metacognitive load, and behavioral inertia converge to predict that students with weaker prior achievement will engage the agent less intensively than their higher-performing peers. These dynamics suppress the intensive use of potentially transformative learning tools and risk compounding existing performance gaps in high-stakes DDD learning environments.

Thus, we hypothesize:

H2: Lower-performing students engage with agentic-AI less frequently than higher-performing students.

2.3. Differential Gains Based on Performance Profile

While prior sections establish a robust positive association between Gen-AI usage and learning outcomes, the question of who benefits most from Gen-AI engagement remains open. The relationship between student performance level and the marginal effectiveness of Gen-AI usage is theoretically ambiguous (Bastani et al. 2024, Nie et al. 2024). On one hand, lower-performing students have greater room for improvement and may benefit disproportionately from Gen-AI’s adaptive scaffolding, targeted feedback, and real-time clarification (Abbas et al. 2024, Clark 2024). On the other hand, higher-performing students—by virtue of stronger metacognitive regulation, more refined learning strategies, and better-developed cognitive schemas—may be more adept at extracting value from AI interactions (see Prather et al. 2024).

2.3.1 Remedial Gains for Low-Performing Students

Even if lower-performing students engage Gen-AI agents less frequently, each prompt may yield a larger conceptual payoff. Because these learners often begin with wider knowledge gaps, less reliable study routines, and lower academic confidence, a single prompt that elicits a clarifying explanation or worked example can correct a core misconception and unlock subsequent material. By contrast, higher-performing students—already near mastery—typically use Gen-AI for fine-tuning: summarizing, editing, or polishing ideas they largely understand. As a result, additional prompts may yield only a modest incremental benefit.

This asymmetry aligns with Vygotsky’s Zone of Proximal Development, which posits that learning accelerates when support targets skills just beyond current competence (Vygotsky 1978). Because lower performing students sit farther below benchmark understanding, their “zone” is wider, giving scaffolding more conceptual ground to cover (Wood et al. 1976, Van de et al. 2010). Gen-AI agents, which diagnose misunderstandings in real time, can tailor feedback precisely to those gaps. Classic studies of intelligent-tutoring systems show that low-proficiency learners reap the largest gains from personalized scaffolding (Bloom 1984, Hattie and Timperley 2007), and recent meta-analyses suggest that Gen-AI interventions

produce especially strong effects for participants with "low prior knowledge" (Wang and Fan 2025).

These scaffolded interactions engage the very cognitive and metacognitive processes that lower-performing students most need to strengthen. Such students often struggle with self-regulated learning behaviors such as goal setting, strategic help-seeking, and sustained persistence (Zimmerman 2002). Immediate, targeted feedback prevents misconceptions from ossifying (Shute 2008) and mirrors the feedback-rich cycles of deliberate practice that builds skill incrementally (Ericsson et al. 1993). Evidence from AI-assisted writing, coding, and statistics contexts, shows that learners with lower baseline skills achieve the greatest score improvements, even with modest overall usage (Kim et al. 2025, Prather et al. 2024).

Taken together, greater potential for foundational gains, feedback that targets knowledge gaps, and motivational reinforcement suggest that Gen-AI engagement may yield stronger learning effects for lower-performing students. Therefore, we hypothesize:

H3a: The positive impact of agentic-AI usage on learning (as measured by performance on the closed-book course final exam) is stronger for low-performing students than for high-performing students.

2.3.2 An Alternative Perspective: Optimization Gains for High-Performing Students

An alternate theoretical perspective suggests that high-performing students may derive greater benefit from Gen-AI agent engagement. The Matthew Effect in education posits that students who start ahead tend to benefit more from learning tools, thereby widening achievement gaps (Ceci and Papierno 2005, Merton 1968, Stanovich 1986, Walberg and Tsai 1983). In graduate DDD courses, high-performing students arrive with richer schemas, sharper metacognition, and more strategic learning habits, positioning them to harvest maximum value from Gen-AI interactions.

High performers' well-organized knowledge structures lower intrinsic cognitive load when they parse Gen-AI feedback (Sweller 1988, Kalyuga 2007), allowing the agent to function as a cognitive amplifier rather than a source of confusion. This cognitive efficiency enables them to integrate new information seamlessly into existing mental models. Students with underdeveloped schemas, by contrast, must devote scarce working-memory resources simply to decode AI output, shrinking the marginal benefit

of each prompt (Kirschner et al. 2006). Educational psychology research confirms that students with well-developed metacognitive skills are more adept at identifying knowledge gaps and using feedback strategically (Azevedo et al. 2013, Pintrich 2002, Winne and Hadwin 1998, Zimmerman 2008).

This alternative perspective suggests that high-performing students derive greater marginal benefit from Gen-AI use through multiple reinforcing mechanisms. Their lower cognitive load, sharper metacognition, higher-order engagement patterns, disciplined study habits, and motivational readiness jointly enable them to extract more actionable insight from every Gen-AI exchange. These advantages may allow high performers to deepen conceptual understanding and accelerate performance in complex, high-stakes educational environments like DDD courses. Therefore, we propose a competing hypothesis,

H3b: The positive impact of agentic-AI agent usage on learning (as measured by performance on the closed-book course final exam) is stronger for high-performing students than for low-performing students.

3. Empirical Context, Data, Analyses, and Results

In this section, we discuss the empirical context, data, model specification, as well as the key findings.

3.1. Empirical Context

To test our hypotheses, we developed and deployed a no-code agentic-AI learning support system—in the form of a virtual data scientist (henceforth referred to as DataAnalystBot)—for an MBA-level introductory data analytics course. The course takes place in term three of an eight-term, one-year MBA program. The students enrolled in the program had an average of four years of prior work experience. A total of 425 students were enrolled in the course, which ran for three weeks between July and August 2024. The course was taught by three instructors across seven sections. To ensure that students from different sections shared similar experiences, all instructors adopted the same set of teaching materials, the same agentic-AI learning support system, and the same assessment methods.

In the course, students learned foundational data analytics concepts and techniques by working through five realistic business cases. Our technology allowed MBA students to use English as their programming language—rather than R or Python—to solve various machine learning problems, presented

as business cases. These spanned unsupervised learning (e.g., segmenting customers at a bank, recommending products for a grocery store using association rule mining) to binary classification (e.g., detecting cancer) to numeric prediction (e.g., estimating how much will a shopper spend) to natural language processing (e.g., detecting fake reviews). To ensure that students meaningfully engaged with these business cases, they were required to work in randomly assigned teams of four to six members (within the same section) and submit a short executive summary of key analyses and responses to case questions for each case. Students remained in the same teams for the duration of the course and interacted with each other via Slack⁴. At the end of the course, students took a proctored, individual, in-person, paper-and-pen, closed-book final exam. Course grades comprised both individual level components (final exam, in-class participation) and group level components (evaluations on the case write-ups, and peer evaluation).

The DataAnalystBot was available to assist students in the course and was built using OpenAI's Assistant API service.⁵ The DataAnalystBot responded to student questions posted on Slack that were addressed to @DataAnalystBot. All case team members (recall that team members remained the same across all the cases) could view the questions submitted to the DataAnalystBot and its responses within their own team but could not see posts from other teams. To help students familiarize themselves with Slack and the DataAnalystBot, students were provided a practice case—with a fictitious business description and minimal data (three columns and 20 rows)—that was separate from the five business cases mentioned earlier at the start of the course. Slack was configured such that each of the six cases (one practice case and five business cases) had a dedicated channel for each team, ensuring that interactions remained private within teams and that team-level conversations related to each case remained separate.

The DataAnalystBot responded to both general analytics-related questions and case-specific queries, as it was built on Retrieval-Augmented Generation (RAG) architecture (Lewis et al. 2020), illustrated in Figure 1. For example, students could address the DataAnalystBot on Slack to request case-

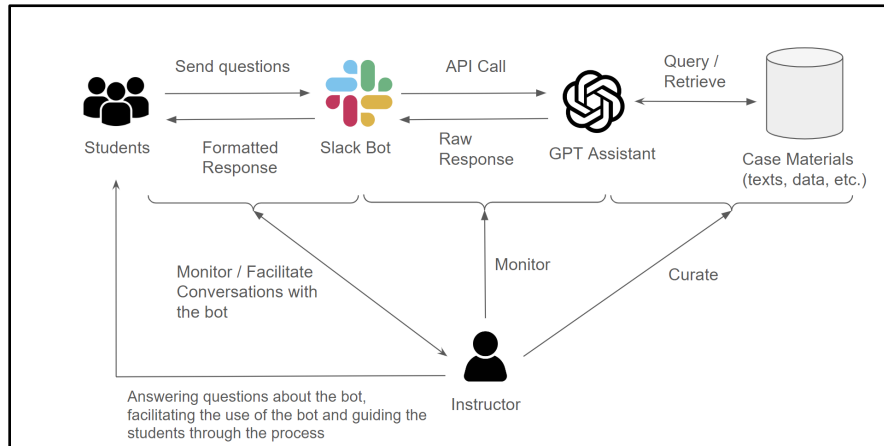
⁴ Slack is an app designed for communication and collaboration, and team discussions.

⁵ We used GPT-4o-2024-05-13, OpenAI's most advanced flagship model at the time of the course (<https://platform.openai.com/docs/assistants/overview>).

specific information (e.g., summarize the business problem); generate and execute code to analyze case datasets (e.g., compute summary statistics, build exploratory or predictive models, and present results); and ask questions related to general data analytics concepts (e.g., explain clustering).

RAG is an AI approach that enhances factual accuracy and reduces hallucinations by integrating real-time information retrieval with language model generation. In other words, instead of relying solely on pre-trained knowledge, RAG first retrieves relevant data from external sources (in our application, it is case materials, texts, datasets, etc., shown on the right of Figure 1) and then incorporates this information into the model's response. This dynamic retrieval process ensures more reliable and contextually accurate outputs, making RAG particularly valuable for applications requiring up-to-date or domain-specific knowledge, as is the case in our context.

Figure 1. DataAnalystBot Architecture



3.2. Data and Variables of Interest

We recorded all interactions between students and the DataAnalystBot throughout the course and aggregated them at the student level to construct the key variables used in subsequent analyses.

Our dependent variables of interest are student learning and frequency of engagement with the agentic-AI tool. We measure the former using each student's final exam score ($exam_i$, where i indexes students). As noted earlier, this was a proctored, individual, paper-and-pen, closed-book (students did not have access to any agentic-AI tools, the Internet or any other assistance) exam, taken in-person. This makes

the final exam score a more reliable indicator of true learning. We captured the frequency of each student's usage of the DataAnalystBot, by measuring the total number of prompts sent to the DataAnalystBot, n_msg_i . We log-transformed this variable to reduce its skewness, i.e., $\log_n_msg_i = \log_2(n_msg_i + 1)$. Our dataset consists of 13,889 student prompts sent to the DataAnalystBot.⁶ $\log_n_msg_i$ also serves as the key independent variable for H1, H3a, and H3b. Another key independent variable is student academic performance. We use each students' GPA from a previous term, ($prev_GPA_i$) to capture intrinsic capability or the ability of a student to earn good grades.⁷

In addition, we incorporated a few important student-level characteristics as control covariates. We controlled for $prompt_length_i$ and $prompt_similarity_i$, which respectively measure the average number of words in student i 's prompts addressed to the DataAnalystBot, and the average semantic similarity between student i 's prompts (also addressed to the DataAnalystBot) and the instructors' demo prompts.⁸ These two variables reflect the amount of effort a student exerts in using the DataAnalystBot. Their inclusion helps alleviate concerns about omitted confounders related to the focal independent variable. Furthermore, we accounted for potential section-level heterogeneity by including section fixed effects, ($section_i$) i.e., a vector of dummy variables corresponding to different sections. Because each instructor was responsible for teaching two to three sections, including section fixed effects also control for instructor-specific heterogeneity. Finally, we controlled for gender with the variable $gender_i$, which takes value 1 if student i is female and 0 otherwise. In Table 1, we present basic summary statistics of the above key variables.

⁶ Since our research question focuses on understanding how usage of the DataAnalystBot affects students' learning outcome, we remove interactions with DataAnalystBot that occurred after the exam. After this exclusion, we are left with 13,889 student prompts.

⁷ More precisely, the course took place during Term 3 of an academic year, and we used students' Term 1 GPA in our analyses. Students took the same set of core courses in Term 1 but could take different electives in Term 2. Therefore, Term 1 GPA is a better reflection of students' pre-existing capabilities than Term 2 GPA.

⁸ During class sessions, the instructors typically demonstrated the use of the DataAnalystBot in each business case with a set of fixed demo prompts. Students are free to follow the demo prompts or write their own prompts. We construct $prompt_similarity_i$ by first deriving the embedding representation of each prompt with Sentence Transformer (Reimers and Gurevych 2019), then computing the average cosine similarity between a student's prompt and the demo prompts within a given business case.

Table 1. Summary Statistics of Key Variables

Variable	Min	Max	Mean	SD
<i>exam</i>	9.00	39.00	29.38	5.16
<i>log_n_msg</i>	0.00	7.23	4.51	1.52
<i>prev_GPA</i>	1.64	4.00	3.25	0.49
<i>prompt_length</i>	0.00	62.48	14.72	6.99
<i>prompt_similarity</i>	0.00	1.00	0.51	0.18
<i>gender</i>	-	-	0.48	0.50

3.3. Main Analyses

We estimate the following regression models to test our hypotheses.

$$exam_i = \beta_0 + \beta_1 log_n_msg_i + \beta_2 prev_GPA_i + \beta_3 prompt_length_i + \beta_4 prompt_similarity_i + \beta_5 gender_i + \Pi section_i + \varepsilon_i \quad (1)$$

$$log_n_msg_i = \beta_0 + \beta_1 prev_GPA_i + \beta_2 prompt_length_i + \beta_3 prompt_similarity_i + \beta_4 gender_i + \Pi section_i + \mu_i \quad (2)$$

$$exam_i = \beta_0 + \beta_1 log_n_msg_i + \beta_2 prev_GPA_i + \beta_3 log_n_msg_i \times prev_GPA_i + \beta_4 prompt_length_i + \beta_5 prompt_similarity_i + \beta_6 gender_i + \Pi section_i + \tau_i \quad (3)$$

Specifically, Model (1) estimates the overall effect of agentic-AI usage on final exam scores, controlling for various student-level characteristics, and corresponds to H1. Model (2) explores the relationship between students' prior academic performance and their agentic-AI usage, corresponding to H2. Model (3) tests for the interaction effect between prior academic performance and agentic-AI usage on final exam scores, corresponding to H3a and H3b. We demean the *prev_GPA_i* variable so that the main effect coefficient in Model (3)—specifically, β_1 —is more interpretable.⁹ Importantly, because students work in teams throughout the course, members of the same team may have correlated agentic-AI usage as well as course performance. To account for such intra-team correlations, we estimate all regression models with

⁹ Without demeaning, β_1 captures the effect of agentic-AI usage on exam scores for a student with a previous GPA of 0. However, the lowest observed GPA in our sample is 1.64. After demeaning, β_1 properly reflects the effect for a student with average prior GPA. Demeaning does not affect the interpretation of coefficients in Models (1) and (2); it only changes the intercept estimates.

standard errors clustered at the team level. Estimation results of these three models are presented in columns (1)-(3) of Table 2.

Table 2. Main Analyses Estimation Results

	(1) DV = exam	(2) DV = log_n_msg	(3) DV = exam
<i>log_n_msg</i>	0.624*** (0.156)		0.606*** (0.155)
<i>prev_GPA</i>	6.972*** (0.468)	0.384*** (0.094)	9.047*** (1.195)
<i>log_n_msg</i> × <i>prev_GPA</i>			-0.484** (0.246)
<i>prompt_length</i>	-0.031 (0.030)	0.039*** (0.010)	-0.033 (0.030)
<i>prompt_similarity</i>	-0.515 (1.385)	4.834*** (0.379)	-0.910 (1.363)
<i>gender</i>	0.450 (0.313)	-0.148 (0.100)	0.382 (0.317)
Constant	26.448** (0.754)	1.231*** (0.232)	26.854*** (0.692)
Section fixed effects	Yes	Yes	Yes
Observations	425	425	425
<i>Adjusted R</i> ²	0.529	0.547	0.533

Notes. Team-level clustered standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Several findings are worth highlighting. Based on column (1), across all students, greater usage of the DataAnalystBot has a positive and statistically significant effect on final exam scores. Thus, we find support for H1. Doubling one's number of prompts sent to the DataAnalystBot is associated with a 0.624 point (or about 1.56 percentage points) increase in final exam score ($p < 0.01$). This effect is economically non-trivial. The final exam is graded on a 0—40 scale and the average score is 29.38. Getting an additional 0.624 points would lift an average student's score percentile from 40.7% to 47.3%, surpassing approximately 28 fellow students. Beyond agentic-AI usage, we find that students with higher prior academic performance also receive significantly higher exam scores. However, the average length of their prompts, their similarity to demo prompts, and gender do not have significant additional effects. Overall, the results from column (1) establish a positive and economically meaningful association between agentic-AI usage and student learning.

Second, we examine results from columns (2)–(3) to understand the moderating role of prior academic performance. Based on column (2), students with higher prior GPA (henceforth referred to as “high performers”) used the DataAnalystBot more than those with lower prior GPA (henceforth referred to as “low performers”).¹⁰ Thus, we find support for H2. A one-point increase in prior GPA is associated with sending approximately 30% more prompts to the DataAnalystBot ($2^{0.384} - 1 = 0.30$). Moreover, column (3) reveals a statistically significant *negative* interaction between agentic-AI usage and prior GPA ($p < 0.05$). Given the positive main effects of agentic-AI usage and prior GPA on final exam scores ($\beta_1 = 0.606$ and $\beta_2 = 9.047$, respectively), this negative interaction implies that low performers benefit *more* from using the DataAnalystBot. Thus, we find support for H3a.

The observed negative interaction effect may stem from the DataAnalystBot helping low performers achieve higher final exam scores, from it somehow hindering high performers, or both. If the DataAnalystBot was hindering high performers, it would raise concerns about the educational value of agentic-AI usage in education. As an empirical test, we split students into “Low Performer” and “High Performer” subsamples, based on whether their prior GPA falls below or above the median value. We then re-estimate regression model (1) excluding the $prev_GPA_i$ term (since it was used to define the subsamples), separately for each group. The results are presented in Table 3.

We find that only the Low Performer subsample (column (1)) shows a positive and statistically significant effect of agentic-AI usage on final exam scores. In the High Performer subsample (column (2)), agentic-AI usage has a positive but statistically non-significant effect. This rules out the possibility that agentic-AI usage harms high performers—an effect that, if present, would have produced a negative coefficient in the High Performer sample. Taken together, the results from Tables 2 and 3 paint an encouraging picture of the academic impact of agentic-AI usage. In our context, the agent functions as an “equalizer”: while high performers engage more with the tool, its usage proves more beneficial for low performers, thereby helping

¹⁰ Table C1 in Appendix C shows student count by prior Performance and DataAnalystBot category (i.e., high vs. low usage).

reduce the performance gap on the final exam. Next, we explore the potential mechanisms underlying this finding.

Table 3. Estimation Results of Low Performer Subsample and High Performer Subsample

	(1) DV = exam Low Performer Sample	(2) DV = exam High Performer Sample
<i>log_n_msg</i>	1.240*** (0.303)	0.294 (0.221)
<i>prompt_length</i>	-0.024 (0.051)	-0.024 (0.046)
<i>prompt_similarity</i>	-2.340 (2.498)	0.639 (1.934)
<i>gender</i>	-0.176 (0.662)	0.010 (0.348)
Constant	23.069*** (1.407)	30.176*** (1.233)
Section fixed effects	Yes	Yes
Observations	201	224
Adjusted R ²	0.153	0.002

Notes. Team-level clustered standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

3.4. Explorations of Underlying Mechanisms

To better understand the equalizing effect of agentic-AI, we conduct a more fine-grained analysis of the different sections of the final exam. Specifically, the exam consists of four components, labeled Q1—Q4, designed to test different skill sets. While Q1—Q3 primarily assess *objective knowledge* of data analytics (e.g., understanding of popular machine learning algorithms and evaluation methods), Q4 is intended to assess students' ability to *apply data analytics techniques to solve problems* (e.g., outline a plan to train, and evaluate a predictive model for target marketing). We decompose each student's overall exam score into *exam_objective_i* (the total score on Q1—Q3) and *exam_apply_i* (the score on Q4). We then estimate regression Model (3) using *exam_objective_i* and *exam_apply_i* as the dependent variables, respectively. The estimation results are reported in Table 4.

We find that agentic-AI usage has a negative and statistically significant interaction with prior GPA on the applied component of the exam, but not the objective component. In other words, the agentic-AI's equalizer effect stems from its ability to help low performers perform better on applied questions.

Table 4. Estimation Results of Agentic-AI Usage across Different Exam Components

	(1) DV = exam objective	(2) DV = exam apply
<i>log_n_msg</i>	0.441*** (0.130)	0.135* (0.078)
<i>prev_GPA</i>	5.780*** (1.044)	3.006*** (0.452)
<i>log_n_msg</i> × <i>prev_GPA</i>	-0.266 (0.217)	-0.170* (0.096)
<i>prompt_length</i>	-0.020 (0.022)	-0.013 (0.014)
<i>prompt_similarity</i>	-0.922 (1.179)	0.176 (0.694)
<i>gender</i>	0.110 (0.252)	0.287* (0.170)
Constant	20.509*** (0.631)	6.243*** (0.350)
Section fixed effects	Yes	Yes
Observations	423	423
Adjusted R ²	0.409	0.440

Notes. Team-level clustered standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. The sample size is 423, as the component-wise final exam scores were not available for two students.

To better understand what type of DataAnalystBot usage contributed to this improvement, we examined how students engaged with the tool during the course. As mentioned earlier, students interacted with the tool both through a practice case and across five actual business cases. The practice case was designed to help students become familiar with the DataAnalystBot. It was introduced at the beginning of the course, and its corresponding channel remained available throughout the course. Interestingly, we observed that students made extensive use of the DataAnalystBot in the practice case channel as a kind of “free ChatGPT,” asking generic questions about technical concepts covered in class and using it to generate mock exam questions (see Appendix A for a few examples). In other words, the DataAnalystBot in the practice case channel unexpectedly served as a “teaching assistant” for the course—one that may have helped low performers *prepare for the final exam*. In contrast, each of the five business cases featured a realistic business problem in data analytics, and students needed to leverage the DataAnalystBot to conduct analyses and develop solutions. As such, usage of the bot across these five business cases could potentially enhance low performers’ *applied problem-solving capability*. To disentangle which of these two types of

agentic-AI usage—usage for final exam preparation (in the practice channel) or usage for applied problem-solving (in the business case channels)—was most beneficial for low performers, we separate students’ prompts sent to DataAnalystBot in the practice channel versus in the five business case channels, and create two new independent variables $\log_n_exam_prep_i$ and $\log_n_case_work_i$ ¹¹ We then re-estimate the regression in Model (2) of Table 4, replacing $\log_n_msg_i$ with these two independent variables.¹² The results are summarized in Table 5.

Table 5. Estimation Results of Agentic-AI Usage in Practice Case vs. Business Cases

	DV = exam_apply
$\log_n_exam_prep$	0.156* (0.080)
$\log_n_case_work$	0.084 (0.079)
$prev_GPA$	2.989*** (0.419)
$\log_n_exam_prep \times prev_GPA$	0.248 (0.192)
$\log_n_case_work \times prev_GPA$	-0.205** (0.094)
$prompt_length$	-0.011 (0.015)
$prompt_similarity$	0.366 (0.749)
$Gender$	0.238 (0.169)
Constant	6.348*** (0.341)
Section fixed effects	Yes
Observations	423
Adjusted R ²	0.447

Notes. Team-level clustered standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. The sample size is 423, as the component-wise final exam scores were not available for two students.

¹¹ Among students’ questions sent to the DataAnalystBot in the practice channel, we removed those related to the agent itself or the interface. The variable $\log_n_exam_prep_i$ includes only questions related to course content.

¹² For completeness, in Appendix B (Table B1) we present the results from Table 5 using $exam_objective$ as the dependent variable. These results show that, for objective exam questions, DataAnalystBot usage (in both practice and business case channels) benefits students with the average prior GPA (the coefficients of $\log_n_exam_prep$ and $\log_n_case_work$ are statistically significant), but there is no “equalizer effect” (as the interaction term coefficients are not statistically significant).

The results support the equalizer effect of agentic-AI usage for low performers, specifically within the actual business cases ($\beta = -0.205, p < 0.05$). This set of analyses reveals that by working on realistic problem-solving tasks with the support of the agentic-AI tool—i.e., what we refer to as the “going to the gym” mode—low performers improved on applied questions of the final exam, allowing them to reduce the gap as compared to high performers.

4. Robustness Checks

In addition to the main analyses above, we perform several robustness checks to assess the validity of our key findings against alternative explanations.

4.1. The “Ceiling Effect”

One alternative explanation for why low performers benefits more from using agentic-AI is that the uplifting effect is simply more detectable for them. In contrast, high performers may face a performance “ceiling”, meaning their final exam scores are already near the upper bound, making marginal improvements from agentic-AI use harder to observe.

We test whether this “ceiling effect” holds water by considering subsamples of students whose exam scores are not extremely close to the ceiling. Specifically, we construct three subsamples consisting of students at or below the 95th, 90th, and 85th percentiles of the final exam score distribution. These subsamples still exhibit meaningful variation in the prior GPA, but students’ final exam scores are no longer tightly constrained by a ceiling.

We re-estimate regression Model (3) from Table 2 on each of these subsamples. The results, shown in Table 6, are consistent with those reported in Model (3) of Table 2. These analyses help alleviate the concerns that our main findings (Table 2, Model (3)) are primarily driven by a ceiling effect among high performers.

Table 6. Estimation Results of Agentic-AI Usage Without Extremely High Performers

	(1) DV = exam (≤ 95% percentile)	(2) DV = exam (≤ 90% percentile)	(3) DV = exam (≤ 85% percentile)
<i>log_n_msg</i>	0.533*** (0.151)	0.498*** (0.155)	0.497*** (0.155)
<i>prev_GPA</i>	9.367*** (1.240)	9.229*** (1.272)	9.204*** (1.258)
<i>log_n_msg</i> × <i>prev_GPA</i>	-0.640** (0.256)	-0.686** (0.265)	-0.707** (0.263)
<i>prompt_length</i>	-0.037 (0.028)	-0.039 (0.028)	-0.035 (0.028)
<i>prompt_similarity</i>	-0.792 (1.362)	-0.623 (1.428)	-0.861 (1.369)
<i>gender</i>	0.522 (0.334)	0.384 (0.334)	0.309 (0.333)
Constant	26.907*** (0.677)	26.801*** (0.674)	26.784*** (0.652)
Section fixed effects	Yes	Yes	Yes
Observations	407	382	375
<i>Adjusted R</i> ²	0.523	0.516	0.514

Notes. Team-level clustered standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

4.2. Observational Learning from Teammates

We further consider another potential mechanism that may account for the greater benefits of agentic-AI use among low performers. Because students work on the business cases in teams, low performers may learn by observing the prompts sent by their high performing teammates. This “observational learning” mechanism offers a plausible alternative explanation for why low performers may reap more of the performance gains even though they use agentic-AI less than high performers.

To test this mechanism, we constructed two team-level variables, $team_GPA_i$ and $log_team_msg_i$. For each student i , these variables capture the average prior GPA of other team-members and the (log-transformed) total number of prompts sent to the DataAnalystBot by the other team-members of their team. Because meaningful teamwork occurs within the five business cases, we focus on the $log_n_case_work_i$ measure as the key independent variable. For the same reason, the $log_team_msg_i$ measure similarly excludes prompts sent to DataAnalystBot in the practice case. With the two team-level

variables incorporated, we re-estimated regression Model (3) from Table 2, with final exam scores as well as final exam scores on the applied questions as dependent variables. The results are presented in Table 7.

Table 7. Estimation Results of Agentic-AI Usage with Team-Level Variables

	(1) DV = exam	(2) DV = exam apply
<i>log_n_case_work</i>	0.550** (0.171)	0.112 (0.304)
<i>prev_GPA</i>	8.857*** (1.089)	3.079*** (0.402)
<i>log_n_case_work</i> × <i>prev_GPA</i>	-0.458* (0.234)	-0.196** (0.090)
<i>team_GPA</i>	-0.214 (0.760)	0.063 (0.358)
<i>log_team_msg</i>	0.286 (0.299)	0.022 (0.161)
<i>prompt_length</i>	-0.030 (0.030)	-0.011 (0.014)
<i>prompt_similarity</i>	-0.678 (1.464)	0.228 (0.740)
<i>gender</i>	0.355 (0.321)	0.271 (0.171)
Constant	-2.913 (5.030)	-4.038** (1.905)
Section fixed effects	Yes	Yes
Observations	425	423
Adjusted R^2	0.528	0.438

Notes. Team-level clustered standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. The sample size in column (2) is 423, as section-level final exam scores are not available for two students.

We find that all key results remain consistent even after incorporating team-level control variables, and none of the team-level variables have significant effects on final exam scores. That is, the positive impact of agentic-AI usage on learning, as well as the equalizer effect for low performers, cannot simply be attributed to “observational learning”. Students—especially low performers—accrue performance benefits from *actively using* the agentic-AI tool rather than by merely observing others.

4.3. Unobserved Heterogeneity in Student Capabilities

Throughout our main analyses, we used students’ prior GPA as the proxy for their learning capabilities. In this robustness check, we further address concern about unobserved heterogeneity in student capabilities by (1) additionally controlling for their GMAT scores and prior work experience and (2) conducting a falsification test.

First, we add two additional control variables, each student's GMAT score ($GMAT_i$) and the number of years of prior work experience in the technology industry ($tech_years_i$).¹³ Both variables reflect students' capabilities prior to entering the program and complement $prev_GPA_i$ in accounting for student-level heterogeneity in learning capabilities.

Second, we conduct a falsification test to further assess whether the above controls sufficiently account for student capability heterogeneity. As discussed earlier, $prev_GPA_i$ is student i 's GPA from Term 1, during which all enrolled students took a common set of core course, whereas the focal course took place during Term 3. In this test, we run a regression with students' GPA at the end of *Term 2* as the dependent variable, while controlling for all the abovementioned variables that proxy student capabilities. Because Term 2 ended before the focal course, students' Term 2 GPA couldn't possibly be affected by their agentic-AI usage. If we have indeed sufficiently accounted for student capability heterogeneity, we would expect agentic-AI usage to show no association (both direct and interaction term) on Term 2 GPA.

We report the estimation results of these two checks in Table 8. Because not all students reported a GMAT score, the sample size of this robustness check decreases from 425 to 342. Column (1) shows that all of our main findings remain robust after the inclusion of the two additional control variables. The falsification test results in column (2), meanwhile, provide further support that our model specification sufficiently controls for heterogeneity in student capabilities.

¹³ We also obtain consistent results using students' GRE scores. However, because fewer students reported GRE scores than GMAT scores, the sample size for this analysis is relatively small.

Table 8. Estimation Results of Agentic-AI Usage with Additional Controls and Falsification Test

	(1) DV = exam	(2) DV = Term 2 GPA
<i>log_n_msg</i>	0.702*** (0.167)	0.019 (0.013)
<i>prev_GPA</i>	11.834*** (1.826)	0.863*** (0.136)
<i>log_n_msg</i> × <i>prev_GPA</i>	-1.058*** (0.333)	-0.042 (0.027)
<i>prompt_length</i>	-0.031 (0.035)	0.002 (0.003)
<i>prompt_similarity</i>	0.086 (1.554)	0.152 (0.096)
<i>gender</i>	0.344 (0.385)	-0.002 (0.027)
<i>GMAT</i>	0.0001 (0.001)	-0.0001 (0.0001)
<i>tech_years</i>	0.094 (0.111)	0.001 (0.006)
Constant	25.810*** (1.211)	3.109*** (0.088)
Section fixed effects	Yes	Yes
Observations	342	342
Adjusted R ²	0.468	0.580

Notes. Team-level clustered standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
The sample size is 342, as data on *GMAT* and *tech_year* variables are available for only 342 students.

5. Discussion

This study examines whether and how agentic-AI tools—specifically, a no-code virtual data scientist—can enhance analytics skill development among MBA students in a high-stakes core analytics course. As data-driven decision making becomes central to modern management practice, developing analytics fluency is critical for MBAs seeking to lead in digitally transformed organizations. We find that higher engagement with an agentic-AI tool is positively associated with learning outcomes, as measured by final exam performance. However, this average effect masks significant heterogeneity: while lower-performing students engage the tool less frequently, they benefit more when they do engage—particularly when applying analytics techniques to open-ended business cases that require both technical modeling and

managerial interpretation. These results illuminate a central paradox in AI-enhanced learning: students who stand to gain the most from agentic-AI support are also the least likely to engage it fully.

Our study advances the burgeoning literature on AI in education in several ways. First, while prior studies, to our knowledge, have all adopted binary “access vs. no access” or single-dose interventions (e.g., Vuculescu et al. 2024, Urban et al. 2024, Adelegan 2024), we move beyond this coarse categorization. We observe the number of queries each student submits across the full duration of an intensive MBA analytics course. This allows us to model final exam performance as a continuous function of engagement intensity, isolating dosage effects that prior studies conflate with mere access.

Second, prior studies report mixed findings on whether GenAI narrows or widens performance gaps—some find equalizing effects (e.g., Vuculescu et al. 2024), while others suggest it amplifies disparities (e.g., Roldán-Monés 2024, Weeks et al. 2024). However, these studies rely on subgroup averages or post-hoc splits, without testing underlying mechanisms. Our study reconciles these differences by explicitly evaluating two dynamics within a unified framework: (a) whether lower-performing students engage with agentic-AI less frequently (a usage gap), and (b) whether, when they do engage, lower- or higher-performing students experience greater learning gains (a returns gap). This dual-moderator design allows us to move beyond descriptive comparisons and pinpoint *when and for whom* agentic-AI usage is most beneficial—revealing that, under certain usage patterns, lower-performing students can meaningfully “level up.”

Third, most existing studies focus on K–12 students (Bastani et al. 2024), STEM lab environments (Rudniy 2024), or general writing courses (Poole and Coss 2024). We extend this literature to a graduate-level, decision-analytics course designed to build managerial data fluency. Our setting—a required MBA course in which final grades significantly influence students’ eligibility for interviews with top consulting firms—introduces real career stakes. This high-stakes, professional-skills context offers a stronger test of agentic-AI’s relevance for future business leaders than prior studies conducted in low-stakes or generic learning environments.

Fourth, prior research often relies on coarse or indirect measures of agentic-AI usage, such as binary indicators of whether students used agentic-AI at all (Tsekouras et al. 2024). Moreover, the few attempts to assess agentic-AI engagement often depend on crude proxies, such as self-reports or usage detection flags from agentic-AI detection systems (Wecks et al. 2024). By contrast, we use detailed prompt-level logs to capture not only how often students used the agent but also how they used it, distinguishing prompts used for solving business cases from those used for exam preparation. This granularity allows us to identify the type of engagement most strongly associated with learning gains. In particular, we find that the largest benefits occur when students use the tool in a “going to the gym” mode—actively working through complex, ill-structured case problems rather than passively reviewing concepts.

Fifth, unlike prior work in higher education that often evaluates outcomes while students still have access to agentic-AI tools (Tsekouras et al. 2024), we measure learning through a closed-book, proctored final exam taken without agentic-AI or other assistance. This design isolates internalized learning gains rather than AI-augmented task performance, providing more credible evidence of durable knowledge acquisition.

Additionally, our study points to several promising directions for future investigation. While our findings suggest that access to AI tools alone is insufficient to ensure equitable learning outcomes, they also show that engagement can disproportionately benefit lower-performing students. In our setting, these students were less likely to engage with the agentic-AI, despite benefiting more when they did—highlighting that behavioral frictions may limit uptake among those who stand to gain the most. This usage–benefit asymmetry points to a need for deeper understanding of how tool design and course structures influence engagement, particularly for students with lower confidence or fluency. Further, future work should examine whether similar usage–benefit asymmetries arise in other domains. For instance, future studies could examine simulation-based clinical courses in nursing education, where students work with simulated patients in hands-on settings while optionally consulting AI tools via mobile devices. These

settings differ in key ways: learning is hands-on and multimodal, combining judgment with the physical execution of nursing procedures; and agentic-AI tools are accessed while alternating between AI use and manual patient care tasks. Clarifying when and how agentic-AI tools equalize rather than exacerbate performance gaps will be essential to guiding their responsible use in diverse educational settings. Nonetheless, our results suggest that thoughtfully embedded agentic-AI tools can serve not only as productivity enhancers but as learning equalizers—if and when engagement gaps are addressed. In this way, the promise of agentic-AI in education lies not only in its technical capability, but in its integration into pedagogical and behavioral systems that shape student learning.

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Appendix A

Example Prompts in the Practice Case

Table A1 collects a few examples to illustrate students' usage of the Gen-AI agent in practice case for exam preparation.

Table A1. Example Prompts in Practice Case

Purpose	Example Prompts
Clarifying Technical Concepts	What is supervised learning and unsupervised learning?
	Is it always the case that the neural network or boosted tree models outperform Decision Tree models?
	How is ROC plot formed? Which is the ideal plot and why? Is it dependent on sensitivity and specificity?
Generate Mock Exam Questions	Prepare 50 questions of all 4 types MCQ, multiple answer, true false and short answer for the below topics in a downloadable format ... <i>[further details omitted for brevity]</i> .
	Prepare more such MCQs for me to practice.
	Now create 20 questions based on the concepts mentioned in Clustering ... <i>[further details omitted for brevity]</i> .

Appendix B

**Table B1. Estimation Results of Gen-AI Usage in Practice Case vs. Business Cases
(DV: exam_objective)**

	DV = exam_objective
<i>log_n_exam_prep</i>	0.325*** (0.104)
<i>log_n_case_work</i>	0.333** (0.139)
<i>prev_GPA</i>	5.474*** (0.995)
<i>log_n_exam_prep</i> × <i>prev_GPA</i>	0.045 (0.226)
<i>log_n_case_work</i> × <i>prev_GPA</i>	-0.213 (0.218)
<i>prompt_length</i>	-0.019 (0.022)
<i>prompt_similarity</i>	-0.445 (1.272)
<i>gender</i>	0.071 (0.257)
Constant	20.653*** (0.633)
Section fixed effects	Yes
Observations	423
Adjusted R ²	0.411

Appendix C

Table C1. Student Count by Prior Performance and DataAnalystBot Usage Category

	Low Performer	High Performer
Low Usage	114	96
High Usage	87	128